第14次习题课2026-6-12, 周五晚7:20 六教6A307(董),6A407(阳)

学习单调函数、绝对连续函数、有界变差函数.

习题1. 设实值函数f在有界闭区间$[a,b]$上连续且单调不减, $E=\{x\in[a,b]\mid f\ \text{在点}x\text{可微}\}$. 证明$f(E)$是L-可测集且

$$m(f(E))=\int_E f'(x)\mathrm{d}x=\int_a^b f'(x)\mathrm{d}x.$$

【证】由Lebesgue定理知f在$[a,b]$上几乎处处可微知E是满测度的, 即$[a,b]\setminus E$是m-零测集, 所以E是L-可测集且

$$\int_E f'(x)\mathrm{d}x=\int_a^b f'(x)\mathrm{d}x.$$

据本讲义**命题7.4.2**知$f(E)$也是L-可测集且(注意$f'(x)\geq 0$)

$$m(f(E))\leq\int_E f'(x)\mathrm{d}x.$$

下证反向不等式也成立. 对任意$\varepsilon>0$, 存在开集$G\supset f(E)$使得$m(G)<m(f(E))+\varepsilon$. 由连续函数的开集刻画和(a,b)是开集知集合

$$A=\{x\in(a,b)\mid f(x)\in G\}\ \text{是开集且易见}\ f(A)\subset G.$$

由$E\subset[a,b]$的m-测度>0知$(a,b)\cap E$非空, 同时显然有$(a,b)\cap E\subset A$, 所以A是非空开集. 所以可写$A=\bigcup\limits_{k\geq1}(a_k,b_k)$, 其中$(a_k,b_k)$ 互不相交. 由$A\subset(a,b)$ 知$[a_k,b_k]\subset[a,b]$. 因此

$$\begin{aligned}&\int_E f'(x)\mathrm{d}x=\int_{(a,b)\cap E}f'(x)\mathrm{d}x\leq\int_A f'(x)\mathrm{d}x=\sum_{k\geq1}\int_{(a_k,b_k)}f'(x)\mathrm{d}x\\&=\sum_{k\geq1}\int_{a_k}^{b_k}f'(x)\mathrm{d}x\leq\sum_{k\geq1}(f(b_k)-f(a_k))=\sum_{k\geq1}m(f([a_k,b_k]))=\sum_{k\geq1}m\big(f((a_k,b_k))\big).\end{aligned}$$

这里用到: f在$[a,b]$上连续且单调不减, 因此$[f(a_k),f(b_k)]=f([a_k,b_k])=f((a_k,b_k))\cup\{f(a_k),f(b_k)\}$, 而$\{f(a_k),f(b_k)\}$是$m$-零测集.

由f单调不减易见$f((a_k,b_k))$ 互不重叠, 即当$i\neq j$时$m(f((a_i,b_i))\cap f((a_j,b_j)))=0$ [否则, $m(f((a_i,b_i))\cap f((a_j,b_j)))>0$. 因$(a_i,b_i),(a_j,b_j)$ 不相交, 故$a_i\neq a_j$. 设$a_i<a_j$, 则$b_i\leq a_j$. 对任意$y\in f((a_i,b_i))\cap f((a_j,b_j))$, 存在$x_i\in(a_i,b_i),x_j\in(a_j,b_j)$ 使得$y=f(x_i)=f(x_j)$. 而由$x_i<b_i\leq a_j<x_j$ 和f单调不减得知$y=f(x_i)\leq f(b_i)\leq f(a_j)\leq f(x_j)=y$, 因此$y=f(b_i)=f(a_j)$, 这表明$f((a_i,b_i))\cap f((a_j,b_j))$是单点集, 与$m(f((a_i,b_i))\cap f((a_j,b_j)))>0$矛盾.] 于是由关于互不重叠的可测集的可加性有

$$\begin{aligned}&\sum_{k\geq1}m\big(f((a_k,b_k))\big)=m\big(\bigcup_{k\geq1}f((a_k,b_k))\big)=m\big(f\big(\bigcup_{k\geq1}(a_k,b_k)\big)\big)\\&=m(f(A))\leq m(G)<m(f(E))+\varepsilon.\end{aligned}$$

所以

$$\int_E f'(x)\mathrm{d}x \le m(f(E)) + \varepsilon.$$

由$\varepsilon > 0$的任意性即得反向不等式

$$\int_E f'(x)\mathrm{d}x \le m(f(E)).$$

□

习题2. 设函数$F : \mathbb{R} \to \mathbb{R}$ 有界、单调不减、且右连续. 由讲义**命题5.5.1** 知在$\mathbb{R}$上存在唯一的正则的Borel测度μ 使得$\mu((a,b]) = F(b) - F(a)$ for all $a < b$. 证明:

$$F\text{在}\mathbb{R}\text{上绝对连续} \iff \mu << m$$

后者表示μ 关于Lebesgue测度m是绝对连续的.

[**注意**: 若将Lebesgue测度m 限制在Borelσ-代数$\mathcal{B}(\mathbb{R})$上, 则m当然也是Borel测度.]

【证】首先由假设和Lebesgue定理知$f(x) := F'(x) \ge 0$ a.e. $x \in \mathbb{R}$ 且

$$\int_{-n}^{n} f(t)\mathrm{d}t \le F(n) - F(-n),$$
$$\int_{-\infty}^{\infty} f(t)\mathrm{d}t \le \lim_{n\to\infty}(F(n) - F(-n)) = F(\infty) - F(-\infty) < \infty.$$

这表明$0 \le f \in L^1(\mathbb{R})$. 同时还有

$$\mu(\mathbb{R}) = \lim_{n\to\infty} \mu((-n,n]) = \lim_{n\to\infty}(F(n) - F(-n)) = F(\infty) - F(-\infty) < \infty.$$

这表明μ是一个有限测度.

现在设F在$\mathbb{R}$上绝对连续. 此时有

$$F(x) = F(-\infty) + \int_{-\infty}^{x} f(t)\mathrm{d}t, \quad x \in \mathbb{R}.$$

对任意开集$G \subset \mathbb{R}$, 由开集的2-进区间分解定理知可写$G = \bigcup_{k=1}^{\infty}(a_k, b_k]$ 其中$(a_k, b_k]$是互不相交的左开右闭的2-进区间. 据此有

$$\mu(G) = \sum_{k=1}^{\infty} \mu((a_k,b_k]) = \sum_{k=1}^{\infty}(F(b_k) - F(a_k)) = \sum_{k=1}^{\infty}\int_{a_k}^{b_k} f(t)\mathrm{d}t = \sum_{k=1}^{\infty}\int_{(a_k,b_k]} f(t)\mathrm{d}t$$
$$= \int_{\bigcup_{k=1}^{\infty}(a_k,b_k]} f(t)\mathrm{d}t = \int_G f(t)\mathrm{d}t.$$

对任意$\varepsilon > 0$, 由$f \in L^1(\mathbb{R})$ 和积分的绝对连续性知, 存在$\delta > 0$ 使得对任意Borel集$E \subset \mathbb{R}$ 满足$m(E) < \delta$ 者都有$\int_E f(t)\mathrm{d}t < \varepsilon$. 取一列递减的开集$G_n \supset E$ 使得$E \cup Z = \bigcap_{n=1}^{\infty} G_n$, 其中$m(Z) = 0$. 则仍由$f$的积分的绝对连续性有

$$\lim_{n\to\infty}\int_{G_n} f(t)\mathrm{d}t = \int_E f(t)\mathrm{d}t < \varepsilon.$$

于是得到

$$\mu(E) \le \lim_{n\to\infty} \mu(G_n) = \lim_{n\to\infty} \int_{G_n} f(t)\mathrm{d}t < \varepsilon.$$

因μ是有限测度, 故由讲义**命题5.5.2** 知$\mu << m$.

反过来, 设$\mu << m$. 则由μ是有限测度和讲义**命题5.5.2** 知对任意$\varepsilon > 0$, 存在$\delta > 0$ 使得对任意Borel集$E \subset \mathbb{R}$ 满足$m(E) < \delta$ 者都有$\mu(E) < \varepsilon$. 注意, 这特别蕴含对任意满足$m(Z) = 0$的Borel集Z 都有$\mu(Z) = 0$. 特别可知单点集是μ-零测集**:**

$$\mu(\{x\}) = 0 \quad \forall\, x \in \mathbb{R}. \qquad \text{这蕴含} \quad \mu((a,b]) = \mu((a,b)).$$

今任取互不相交的开区间$(a_k, b_k), k = 1, 2, ..., n$, 满足$\mu_{k=1}^{n}(b_k - a_k) < \delta$. 则有$m(\bigcup_{k=1}^{n}(a_k, b_k)) = \sum_{k=1}^{n}(b_k - a_k) < \delta$ 从而有$\mu(\bigcup_{k=1}^{n}(a_k, b_k)) < \varepsilon$, 因此

$$\sum_{k=1}^{n} |F(b_k) - F(a_k)| = \sum_{k=1}^{n} \mu((a_k, b_k]) = \sum_{k=1}^{n} \mu((a_k, b_k)) = \mu(\bigcup_{k=1}^{n}(a_k, b_k)) < \varepsilon.$$

所以F在$\mathbb{R}$上绝对连续. □

习题3. 设f_n在$[a,b]$上单调不减且一致有界: $A \le f_n(x) \le B \quad \forall\, x \in [a,b], n = 1, 2, 3, \dots$, 其中$-\infty < A < B < \infty$. 令

$$F(x) = \sum_{n=1}^{\infty} 2^{-n} f_n(x), \quad x \in [a,b].$$

证明F在$[a,b]$上绝对连续$\Longleftrightarrow$ 每个f_n在$[a,b]$上绝对连续.

【证】首先由Fubini逐项微分定理知

$$F'(x) = \sum_{n=1}^{\infty} 2^{-n} f_n'(x) \qquad \text{a.e.} \quad x \in [a,b].$$

因每个$f_n'(x)$都是L-非负可测函数, 故成立逐项积分:

$$\int_a^b F'(x)\mathrm{d}x = \sum_{n=1}^{\infty} 2^{-n} \int_a^b f_n'(x)\mathrm{d}x.$$

又据Lebesgue定理知(注意f_n都是单调不减的)

$$\int_a^b f_n'(x)\mathrm{d}x \le f_n(b) - f_n(a), \quad n = 1, 2, 3, \dots.$$

于是由关于单调函数成为绝对连续函数的一个充要条件(见**§7.2**的习题**7.2.4**) 即知

$$\begin{aligned} &F\text{在}[a,b]\text{上绝对连续} \Longleftrightarrow \int_a^b F'(x)\mathrm{d}x = F(b) - F(a) \\ &\Longleftrightarrow \sum_{n=1}^{\infty} 2^{-n} \int_a^b f_n'(x)\mathrm{d}x = \sum_{n=1}^{\infty} 2^{-n}\big(f_n(b) - f_n(a)\big) \\ &\Longleftrightarrow \sum_{n=1}^{\infty} 2^{-n}\big(f_n(b) - f_n(a) - \int_a^b f_n'(x)\mathrm{d}x\big) = 0 \end{aligned}$$

$$\Longleftrightarrow f_n(b)-f_n(a)-\int_a^b f_n'(x)\mathrm{d}x=0,\quad n=1,2,3,...;$$
$$\Longleftrightarrow \int_a^b f_n'(x)\mathrm{d}x=f_n(b)-f_n(a),\quad n=1,2,3,...;$$
$\Longleftrightarrow$ 每个f_n在$[a,b]$上绝对连续.

□

习题4. 试在$\mathbb{R}$上构造一个连续且严格单调增加的函数f 满足$f'(x)=0$ a.e. $x\in\mathbb{R}$.

【解】设$g(x)$是$[0,1]$上的通过零测度Cantor集C_0构作的Cantor函数. 回忆Cantor集C_0: $[0,1]\setminus C_0=\bigcup_{k=1}^{\infty}(a_k,b_k)$, 其中开区间$(a_k,b_k)\subset(0,1)$互不相交且$\sum_{k=1}^{\infty}(b_k-a_k)=1$. 而Cantor函数$g$在$[0,1]$上连续且单调不减, $g(0)=0,g(1)=1$, 且g在每个开区间(a_k,b_k)上是常数, 因此$g'(x)=0$ a.e. $x\in[0,1]$.

现将g连续延拓到$\mathbb{R}$: $g(x)=0$ for all $x\le 0$; $g(x)=1$ for all $x\ge 1$. 则延拓后的Cantor函数g在$\mathbb{R}$上连续且单调不增, 且$g'(x)=0$ for all $x\in\mathbb{R}\setminus Z_0$ 其中$m(Z_0)=0$.

考虑可数集$A:=\{(a,b)\in\mathbb{Q}\times\mathbb{Q}\mid a<b\}$, 将其写成$A=\{(a_n,b_n)\}_{n=1}^{\infty}$. 令

$$Z_n=a_n+(b_n-a_n)Z_0,\quad Z=\bigcup_{n=1}^{\infty}Z_n,$$
$$f_n(x)=g\big(\frac{x-a_n}{b_n-a_n}\big),\quad x\in\mathbb{R}.$$

则易见$m(Z_n)=(b_n-a_n)m(Z_0)=0, n=1,2,3,...$, 因此$m(Z)=0$. 对任意$x\in\mathbb{R}\setminus Z$, 对任意$n\in\mathbb{N}$, 有$x\in\mathbb{R}\setminus Z_n$ 即$\frac{x-a_n}{b_n-a_n}\in\mathbb{R}\setminus Z_0$, 因此$g$ 在点$\frac{x-a_n}{b_n-a_n}$可导且$g'(\frac{x-a_n}{b_n-a_n})=0$, 因此由复合函数求导法则知$f_n$在$x$可导且

$$f_n'(x)=g'\big(\frac{x-a_n}{b_n-a_n}\big)\frac{1}{b_n-a_n}=0.$$

这证明了

$$f_n'(x)=0\qquad\forall\,x\in\mathbb{R}\setminus Z,\quad\forall\,n=1,2,3,\dots.$$

令

$$f(x)=\sum_{n=1}^{\infty}2^{-n}f_n(x),\quad x\in\mathbb{R}.$$

则由$0\le f_n\le 1$知这是一个一致收敛的函数级数, 因此其和函数f在$\mathbb{R}$ 上连续且单调不减. 由Fubini逐项微分定理和$f_n'(x)=0$ a.e. $x\in\mathbb{R}$ 得知

$$f'(x)=\sum_{n=1}^{\infty}2^{-n}f_n'(x)=0\qquad\text{a.e.}\quad x\in\mathbb{R}.$$

另一方面, 对任意$x,y\in\mathbb{R}, x<y$, 由有理数的稠密性, 存在有理数a,b 使得$x<a<b<y$. 由A的定义知$(a,b)\in A$, 因此存在$k\in\mathbb{N}$使得$(a,b)=(a_k,b_k)$, 从而有$x<a_k<b_k<y$.于是由f_n单调不减有

$$f(y)-f(x)=\sum_{n=1}^{\infty}2^{-n}\big(f_n(y)-f_n(x)\big)\ge 2^{-k}\big(f_k(y)-f_k(x)\big)$$

$$\geq 2^{-k}\big(f_k(b_k)-f_k(a_k)\big)=2^{-k}\big(g(1)-g(0)\big)=2^{-k}>0.$$

这表明f在$\mathbb{R}$上是严格单调增加的. 函数f的这个奇异性立即蕴含f在任何闭子区间$[a,b]$上都不是绝对连续的: 我们有

$$\int_a^b f'(x)\mathrm{d}x=0<f(b)-f(a).$$

□

习题5(有界变差函数的一个等价定义). 设$I\subset\mathbb{R}$为开区间, 实值函数$f\in L^1_{loc}(I)$. 证明: f, 或f经过在I中的一个m-零测集上修正后, 满足在I中的每个有界闭子区间上都是有界变差的 $\Longleftrightarrow$ 对任意有界闭区间$[a,b]\subset I$, 存在常数$0\leq A<\infty, 0<\delta<\min\{a-\inf I,\sup I-b\}$ 使得

$$\int_a^b|f(x+h)-f(x)|\mathrm{d}x\leq A|h|\qquad\forall\,|h|\leq\delta.$$

[注: 若f已经在I上连续, 则对f不需要修正.]

【证】"$\Longrightarrow$": 任取有界闭区间$[a,b]\subset I$. 取$0<\delta<\min\{a-\inf I,\sup I-b\}$. 考虑变差函数

$$G(x)=\mathrm{V}_{a-\delta}^{x}(f),\quad x\in[a-\delta,b+\delta];\qquad A=G(b+\delta)=\mathrm{V}_{a-\delta}^{b+\delta}(f).$$

则$0\leq A<\infty$ 且对任意$|h|\leq\delta$, 当$h\geq 0$时有(**应用变差的可加性**)

$$\begin{aligned}&\int_a^b|f(x+h)-f(x)|\mathrm{d}x\leq\int_a^b\mathrm{V}_x^{x+h}(f)\mathrm{d}x=\int_a^b(G(x+h)-G(x))\mathrm{d}x\\&=\int_a^bG(x+h)\mathrm{d}x-\int_a^bG(x)\mathrm{d}x=\int_{a+h}^{b+h}G(x)\mathrm{d}x-\int_a^bG(x)\mathrm{d}x\\&=\int_b^{b+h}G(x)\mathrm{d}x-\int_a^{a+h}G(x)\mathrm{d}x\leq G(b+h)h\leq A|h|,\end{aligned}$$

当$h\leq 0$时,

$$\begin{aligned}&\int_a^b|f(x+h)-f(x)|\mathrm{d}x\leq\int_a^b\mathrm{V}_{x+h}^{x}(f)\mathrm{d}x=\int_a^b(G(x)-G(x+h))\mathrm{d}x\\&=\int_a^bG(x)\mathrm{d}x-\int_a^bG(x+h)\mathrm{d}x=\int_a^bG(x)\mathrm{d}x-\int_{a+h}^{b+h}G(x)\mathrm{d}x\\&=\int_{b+h}^bG(x)\mathrm{d}x-\int_{a+h}^aG(x)\mathrm{d}x\leq G(b)(-h)\leq A|h|.\end{aligned}$$

"$\Longleftarrow$": 对任意$n\in\mathbb{N}, x\in I$, 令

$$f_n(x)=n\int_x^{x+1/n}f(t)\mathrm{d}t\ \text{ 若}[x,x+1/n]\subset I;\quad f_n(x)=0\ \text{ 若}[x,x+1/n]\not\subset I.$$

由假设, 若$[x,x+1/n]\subset I$, 则f在$[x,x+1/n]$上可积, 所以$f_n(x)$ 有定义.

因I是开区间, 故对每个$x \in I$, 当$n >> 1$时有$[x, x+1/n] \subset I$. 因此再由$L^1_{loc}(I)$和Lebesgue微分定理知存在一个m-零测集$Z \subset I$, 使得

$$\lim_{n\to\infty} f_n(x) = f(x) \qquad \forall\, x \in I \setminus Z. \tag{1}$$

取定任一有界闭区间$[a, b] \subset I$. 取定$b < b^* < \sup I$. 由假设, 存在$0 \le A < \infty, 0 < \delta < \sup I - b^*$ 使得

$$\int_a^{b^*} |f(x+h) - f(x)|\mathrm{d}x \le Ah \qquad \forall\, h \in [0, \delta]$$

设$n \in \mathbb{N}$ 使得$1/n < \delta$. 则当$x \in [a, b^*]$时$[x, x+1/n] \subset I$. 因此对任意$x, y \in [a, b^*]$ 有

$$\begin{aligned} f_n(y) - f_n(x) &= n\int_y^{y+1/n} f(t)\mathrm{d}t - n\int_x^{x+1/n} f(t)\mathrm{d}t \\ &= n\int_y^{x} f(t)\mathrm{d}t + n\int_x^{x+1/n} f(t)\mathrm{d}t + n\int_{x+1/n}^{y+1/n} f(t)\mathrm{d}t - n\int_x^{x+1/n} f(t)\mathrm{d}t \\ &= n\int_{x+1/n}^{y+1/n} f(t)\mathrm{d}t - n\int_x^{y} f(t)\mathrm{d}t = n\int_x^y \big(f(t+1/n) - f(t)\big)\mathrm{d}t \end{aligned}$$

$\Longrightarrow$

$$|f_n(y) - f_n(x)| \le n\int_{x\wedge y}^{x\vee y} |f(t+1/n) - f(t)|\mathrm{d}t, \quad x, y \in [a, b^*] \tag{2}$$

$\Longrightarrow$

$$|f_n(y) - f_n(x)| \le n\int_a^{b^*} |f(t+1/n) - f(t)|\mathrm{d}t \le A, \qquad x, y \in [a, b^*].$$

取定一点$x_0 \in (a, b) \setminus Z$. 则对任意$y \in [a, b^*] \setminus Z$ 有

$$|f_n(y) - f_n(x_0)| \le A.$$

令$n \to \infty$ 由(1)得知

$$|f(y) - f(x_0)| = \lim_{n\to\infty} |f_n(y) - f_n(x_0)| \le A.$$

这证明了f在$[a, b^*] \setminus Z$ 上有界:

$$\sup_{y\in[a,b^*]\setminus Z} |f(y)| \le |f(x_0)| + A < \infty.$$

由$[a, b] \subset I$的任意性知f 在$I \setminus Z$上局部有界. 定义

$$F(x) = \begin{cases} f(x) & \text{若} \quad x \in I \setminus Z, \\ \liminf\limits_{I\setminus Z \ni y \to x+} f(y) & \text{若} \quad x \in Z. \end{cases}$$

这个F是f在零测集Z上的一个修正. 必须注意, f在$I \setminus Z$ 上的局部有界性和m-零测集的余集的稠密性蕴含$F(x)$ 处处有限, 即$F(x) \in \mathbb{R}$ for all $x \in I$.

下面证明: 对任意有界闭区间$[a, b] \subset I$, F在$[a, b]$上是有界变差的.

设$[a, b], b^*$ 和A如上. 对于$[a, b]$的任意分划$a = x_0 < x_1 < x_2 < \cdots < x_N = b$, 令

$$0 < \varepsilon < \min\{x_1 - x_0,\, x_2 - x_1,\, ...,\, x_N - x_{N-1},\, b^* - b\}.$$

对每个分点x_i, 若$x_i \in Z$, 则由F的定义知存在数列$\{x_i^{(k)}\}_{k=1}^{\infty} \subset (x_i, x_i+\varepsilon)\setminus Z$ 使得$f(x_i^{(k)}) \to F(x_i)\,(k\to\infty)$; 若$x_i \notin Z$, 则定义$x_i^{(k)} \equiv x_i$, 此时有$f(x_i^{(k)}) = f(x_i) = F(x_i), k = 1,2,3,....$ 于是有$a \le x_0^{(k)} < x_1^{(k)} < x_2^{(k)} < \cdots < x_N^{(k)} < b^*$ 从而由(2)有

$$\sum_{i=0}^{N-1} |f_n(x_{i+1}^{(k)}) - f_n(x_i^{(k)})| \le n\sum_{i=0}^{N-1}\int_{x_i^{(k)}}^{x_{i+1}^{(k)}} |f(t+1/n)-f(t)|\mathrm{d}t$$
$$= n\int_{x_0^{(k)}}^{x_N^{(k)}} |f(t+1/n)-f(t)|\mathrm{d}t \le n\int_a^{b^*} |f(t+1/n)-f(t)|\mathrm{d}t \le A \qquad \forall\, n \ge 1/\delta.$$

因一切$x_i^{(k)} \in I\setminus Z$, 故由(1)得到

$$\sum_{i=0}^{N-1} |f(x_{i+1}^{(k)}) - f(x_i^{(k)})| = \lim_{n\to\infty}\sum_{k=0}^{N-1} |f_n(x_{i+1}^{(k)}) - f_n(x_i^{(k)})| \le A \qquad \forall\, k \in \mathbb{N}.$$

再令$k\to\infty$ 得到

$$\sum_{i=0}^{N-1} |F(x_{i+1}) - F(x_i)| = \lim_{k\to\infty}\sum_{i=0}^{N-1} |f(x_{i+1}^{(k)}) - f(x_i^{(k)})| \le A.$$

据分划的任意性即知f的这个修正函数F在$[a,b]$上是有界变差的. □

【多元有界变差函数的定义】若实值函数$f \in L^1_{loc}(\mathbb{R}^d)$ 满足

$$\sup_{0\ne h\in\mathbb{R}^d} \frac{1}{|h|}\int_{\mathbb{R}^d} |f(x+h)-f(x)|\mathrm{d}x < \infty$$

则称f在$\mathbb{R}^d$上是一个有界变差函数, 或称f在$\mathbb{R}^d$上是有界变差的. $\mathbb{R}^d$上有界变差的实值函数的全体记作$\mathrm{BV}(\mathbb{R}^d)$.

设$E \subset \mathbb{R}^d$为一个Lebesgue 可测集, $f: E\to\mathbb{R}$ 为一个Lebesgue 可测函数. 若f的零延拓$\mathbf{1}_E f \in \mathrm{BV}(\mathbb{R}^d)$, 则称$f$在$E$上是一个有界变差函数, 或称$f$在$E$上是有界变差的. E上有界变差的实值函数的全体记作$\mathrm{BV}(E)$.

习题6(多元有界变差函数的等价定义). 设实值函数$f \in L^1_{loc}(\mathbb{R}^d)$. 证明

$$f \in \mathrm{BV}(\mathbb{R}^d) \Longleftrightarrow \sup_{\varphi\in C_c^1(\mathbb{R}^d),\|\varphi\|_\infty\le 1} \Big|\int_{\mathbb{R}^d} f(x)\frac{\partial}{\partial x^i}\varphi(x)\mathrm{d}x\Big| < \infty \quad \forall\, i = 1,2,...,d. \qquad (*)$$

这里$\|\varphi\|_\infty = \sup\limits_{x\in\mathbb{R}^d}|\varphi(x)|$.

【证】“$\Longrightarrow$”: 设$f\in\mathrm{BV}(\mathbb{R}^d)$, 令

$$A = \sup_{0\ne h\in\mathbb{R}^d}\frac{1}{|h|}\int_{\mathbb{R}^d}|f(x+h)-f(x)|\mathrm{d}x.$$

任取$\varphi\in C_c^1(\mathbb{R}^d)$, 任取$i\in\{1,2,...,d\}$. 对任意$t\in\mathbb{R}$ 有

$$\int_{\mathbb{R}^d} f(x)\big(\varphi(x+t\mathbf{e}_i)-\varphi(x)\big)\mathrm{d}x = \int_{\mathbb{R}^d} f(x)\varphi(x+t\mathbf{e}_i)\mathrm{d}x - \int_{\mathbb{R}^d} f(x)\varphi(x)\mathrm{d}x$$
$$= \int_{\mathbb{R}^d} f(x-t\mathbf{e}_i)\varphi(x)\mathrm{d}x - \int_{\mathbb{R}^d} f(x)\varphi(x)\mathrm{d}x = \int_{\mathbb{R}^d}\big(f(x-t\mathbf{e}_i)-f(x)\big)\varphi(x)\mathrm{d}x$$

$$\Longrightarrow$$

$$\Big|\int_{\mathbb{R}^d} f(x)\big(\varphi(x+t\mathbf{e}_i)-\varphi(x)\big)\mathrm{d}x\Big| \le \int_{\mathbb{R}^d} \big|f(x-t\mathbf{e}_i)-f(x)\big||\varphi(x)|\mathrm{d}x$$

$$\le \int_{\mathbb{R}^d} \big|f(x-t\mathbf{e}_i)-f(x)\big|\mathrm{d}x \le A|t|$$

$$\Longrightarrow$$

$$\Big|\int_{\mathbb{R}^d} f(x)\frac{\varphi(x+t\mathbf{e}_i)-\varphi(x)}{t}\mathrm{d}x\Big| \le A \qquad \forall\, t\in\mathbb{R}\setminus\{0\}. \tag{$**$}$$

由$\varphi \in C_c^1(\mathbb{R}^d)$知存在$0<R<\infty$ 使得$\varphi(x)=0$ for all $|x|\ge R$. 令$C_i=\big\|\frac{\partial}{\partial x^i}\varphi\big\|_\infty$, 则$0\le C_i<\infty$且

$$\Big|f(x)\frac{\varphi(x+t\mathbf{e}_i)-\varphi(x)}{t}\Big| \le C_i|f(x)|\mathbf{1}_{B(0,R+1)}(x) \quad \forall\, x\in\mathbb{R}^d, t\in\mathbb{R}\setminus\{0\},$$

$$\lim_{t\to 0} f(x)\frac{\varphi(x+t\mathbf{e}_i)-\varphi(x)}{t} = f(x)\frac{\partial}{\partial x^i}\varphi(x) \quad \forall\, x\in\mathbb{R}^d.$$

因此据LDC 和$(**)$ 得到

$$\Big|\int_{\mathbb{R}^d} f(x)\frac{\partial}{\partial x^i}\varphi(x)\mathrm{d}x\Big| = \lim_{t\to 0}\Big|\int_{\mathbb{R}^d} f(x)\frac{\varphi(x+t\mathbf{e}_i)-\varphi(x)}{t}\mathrm{d}x\Big| \le A.$$

由$\varphi\in C_c^1(\mathbb{R}^d)$和$i\in\{1,2,...,d\}$的任意性得知

$$\sup_{\varphi\in C_c^1(\mathbb{R}^d),\|\varphi\|_\infty\le 1}\Big|\int_{\mathbb{R}^d} f(x)\frac{\partial}{\partial x^i}\varphi(x)\mathrm{d}x\Big| \le A<\infty \quad \forall\, i=1,2,...,d.$$

“$\Longleftarrow$”：设f满足$(*)$中右边的条件. 对任意$i\in\{1,2,...,d\}$, 令

$$A_i = \sup_{\varphi\in C_c^1(\mathbb{R}^d),\|\varphi\|_\infty\le 1}\Big|\int_{\mathbb{R}^d} f(x)\frac{\partial}{\partial x^i}\varphi(x)\mathrm{d}x\Big|.$$

任取$h^i\in\mathbb{R}$ 和$R\in(1+|h^i|,\infty)$. 对于任意$0<\delta<1$, 设$0\le\varphi\in C_c^\infty(\mathbb{R}^d)$, $\varphi_\delta(x)=\delta^{-d}\varphi(\delta^{-1}x)$, 是常用的磨光函数, 即$\mathrm{supp}\varphi=\overline{B(0,1)}$, $\int_{\mathbb{R}^d}\varphi(x)\mathrm{d}x=1$, 从而$0\le\varphi_\delta\in C_c^\infty(\mathbb{R}^d)$, $\mathrm{supp}\varphi_\delta=\overline{B(0,\delta)}$, $\int_{\mathbb{R}^d}\varphi_\delta(x)\mathrm{d}x=1$. 考虑卷积逼近

$$f_{\delta,R}(x) := (f_R*\varphi_\delta)(x) = \int_{\mathbb{R}^d} f_R(y)\varphi_\delta(x-y)\mathrm{d}y, \quad x\in\mathbb{R}^d.$$

这里

$$f_R(x) = f(x)\mathbf{1}_{B(0,3R)}(x).$$

因$f_R\in L^1(\mathbb{R}^d)$, 故由卷积逼近定理知

$$\int_{\mathbb{R}^d}|f_R(x)-f_{\delta,R}(x)|\mathrm{d}x\to 0 \quad (\delta\to 0^+).$$

计算

$$f_{\delta,R}(x+h^i\mathbf{e}_i)-f_{\delta,R}(x) = \int_{\mathbb{R}^d} f_R(y)\big(\varphi_\delta(x+h^i\mathbf{e}_i-y)-\varphi_\delta(x-y)\big)\mathrm{d}y$$

$$= \int_{\mathbb{R}^d} f_R(y)\Big(\int_0^1 \big(\frac{\partial}{\partial x^i}\varphi_\delta\big)(x+th^i\mathbf{e}_i-y)h^i\mathrm{d}t\Big)\mathrm{d}y$$

$$= h^i \int_{\mathbb{R}^d} f_R(y)\Big(\int_0^1 (\frac{\partial}{\partial x^i}\varphi_\delta)(x + th^i\mathbf{e}_i - y)\mathrm{d}t\Big)\mathrm{d}y.$$

考虑符号函数(**这是证明中的唯一技术要点**)

$$\psi_i(x) = \psi_{i,R,\delta,h^i}(x) := \mathrm{sgn}\Big(\int_{\mathbb{R}^d} f_R(y)\Big(\int_0^1 (\frac{\partial}{\partial x^i}\varphi_\delta)(x + th^i\mathbf{e}_i - y)\mathrm{d}t\Big)\mathrm{d}y\Big), \quad x \in \mathbb{R}^d.$$

易见ψ_i是$\mathbb{R}^d$上的有界的Borel 可测函数, 且

$$\begin{aligned}
&\Big| \int_{\mathbb{R}^d} f_R(y)\Big(\int_0^1 (\frac{\partial}{\partial x^i}\varphi_\delta)(x + th^i\mathbf{e}_i - y)\mathrm{d}t\Big)\mathrm{d}y\Big| \\
&= \psi_i(x) \int_{\mathbb{R}^d} f_R(y)\Big(\int_0^1 (\frac{\partial}{\partial x^i}\varphi_\delta)(x + th^i\mathbf{e}_i - y)\mathrm{d}t\Big)\mathrm{d}y \\
&= \int_{\mathbb{R}^d} \psi_i(x) f_R(y)\Big(\int_0^1 (\frac{\partial}{\partial x^i}\varphi_\delta)(x + th^i\mathbf{e}_i - y)\mathrm{d}t\Big)\mathrm{d}y.
\end{aligned}$$

因此

$$\begin{aligned}
&|f_{\delta,R}(x + h^i\mathbf{e}_i) - f_{\delta,R}(x)| = |h^i| \int_{\mathbb{R}^d} f_R(y)\Big(\int_0^1 \psi_i(x)\big(\frac{\partial}{\partial x^i}\varphi_\delta\big)(x + th^i\mathbf{e}_i - y)\mathrm{d}t\Big)\mathrm{d}y \\
&\Longrightarrow \\
&\int_{B(0,R)} |f_{\delta,R}(x + h^i\mathbf{e}_i) - f_{\delta,R}(x)|\mathrm{d}x \\
&= |h^i| \int_{\mathbb{R}^d} f_R(y)\Big(\int_0^1 \Big(\int_{B(0,R)} \psi_i(x)\Big(\frac{\partial}{\partial x^i}\varphi_\delta\Big)(x + th^i\mathbf{e}_i - y)\mathrm{d}x\Big)\mathrm{d}t\Big)\mathrm{d}y.
\end{aligned}$$

令

$$\Phi_i(y) = \Phi_{i,R,\delta,h^i}(y) := -\int_0^1 \Big(\int_{B(0,R)} \psi_i(x)\varphi_\delta(x + th^i\mathbf{e}_i - y)\mathrm{d}x\Big)\mathrm{d}t, \quad y \in \mathbb{R}^d.$$

由积分号下求导定理知$\Phi_i \in C^\infty(\mathbb{R}^d)$ 且

$$\begin{aligned}
\frac{\partial}{\partial y^i}\Phi_i(y) &= \int_0^1 \Big(\int_{B(0,R)} \psi_i(x)\Big(-\frac{\partial}{\partial y^i}\big(\varphi_\delta(x + th^i\mathbf{e}_i - y)\big)\Big)\mathrm{d}x\Big)\mathrm{d}t \\
&= \int_0^1 \Big(\int_{B(0,R)} \psi_i(x)\Big(\frac{\partial}{\partial x^i}\varphi_\delta\Big)(x + th^i\mathbf{e}_i - y)\mathrm{d}x\Big)\mathrm{d}t \quad \forall\, y \in \mathbb{R}^d.
\end{aligned}$$

因此

$$\int_{B(0,R)} |f_{\delta,R}(x + h^i\mathbf{e}_i) - f_{\delta,R}(x)|\mathrm{d}x = |h^i| \int_{\mathbb{R}^d} f_R(y)\frac{\partial}{\partial y^i}\Phi_i(y)\mathrm{d}y. \qquad (***)$$

因当$\Phi_i(y) \neq 0$ 时, 存在$x \in B(0,R)$和$t \in [0,1]$使得$\varphi_\delta(x + th^i\mathbf{e}_i - y) \neq 0$ 从而有$|x + th^i\mathbf{e}_i - y| < \delta < 1 \Longrightarrow |y| \leq 1 + |x + th^i\mathbf{e}_i| \leq 1 + |x| + |h^i| < 1 + |h^i| + R < 2R$. 所以$\mathrm{supp}\Phi_{\delta,i} \subset \overline{B(0, 1 + |h^i| + R)} \subset B(0, 2R)$, 所以$\Phi_i \in C_c^\infty(\mathbb{R}^d)$. 同时有

$$\begin{aligned}
|\Phi_i(y)| &\leq \int_0^1 \Big(\int_{B(0,R)} |\psi_i(x)|\varphi_\delta(x + th^i\mathbf{e}_i - y)\mathrm{d}x\Big)\mathrm{d}t \\
&\leq \int_0^1 \Big(\int_{\mathbb{R}^d} \varphi_\delta(x + th^i\mathbf{e}_i - y)\mathrm{d}x\Big)\mathrm{d}t = \int_{\mathbb{R}^d} \varphi_\delta(x)\mathrm{d}x = 1 \qquad \forall\, y \in \mathbb{R}^d.
\end{aligned}$$

所以$\|\Phi_i\|_\infty \le 1$. 易见当$y \notin \overline{B(0,2R)}$时$\frac{\partial}{\partial y^i}\Phi_i(y) = 0$, 所以有

$$f_R(y)\frac{\partial}{\partial y^i}\Phi_i(y) \equiv f(y)\frac{\partial}{\partial y^i}\Phi_i(y), \quad y \in \mathbb{R}^d$$

于是由$(***)$ 和假设(即$(*)$的右边)有

$$\int_{B(0,R)} |f_{\delta,R}(x + h^i\mathbf{e}_i) - f_{\delta,R}(x)|\mathrm{d}x = |h^i| \int_{\mathbb{R}^d} f(y)\frac{\partial}{\partial y^i}\Phi_i(y)\mathrm{d}y \le A_i|h^i|.$$

由此有

$$\begin{aligned}
&\int_{B(0,R)} |f_R(x + h^i\mathbf{e}_i) - f_R(x)|\mathrm{d}x \\
&= \int_{B(0,R)} |f_R(x + h^i\mathbf{e}_i) - f_{\delta,R}(x + h^i\mathbf{e}_i) + f_{\delta,R}(x + h^i\mathbf{e}_i) - f_{\delta,R}(x) + f_{\delta,R}(x) - f_R(x)|\mathrm{d}x \\
&\le \int_{B(0,R)} |f_R(x + h^i\mathbf{e}_i) - f_{\delta,R}(x + h^i\mathbf{e}_i)|\mathrm{d}x + \int_{B(0,R)} |f_{\delta,R}(x + h^i\mathbf{e}_i) - f_{\delta,R}(x)|\mathrm{d}x \\
&+ \int_{B(0,R)} |f_{\delta,R}(x) - f_R(x)|\mathrm{d}x \\
&\le \int_{\mathbb{R}^d} |f_R(x + h^i\mathbf{e}_i) - f_{\delta,R}(x + h^i\mathbf{e}_i)|\mathrm{d}x + A_i|h^i| + \int_{\mathbb{R}^d} |f_{\delta,R}(x) - f_R(x)|\mathrm{d}x \\
&= 2\int_{\mathbb{R}^d} |f_{\delta,R}(x) - f_R(x)|\mathrm{d}x + A_i|h^i| \to A_i|h^i| \quad (\delta \to 0^+).
\end{aligned}$$

所以

$$\int_{B(0,R)} |f_R(x + h^i\mathbf{e}_i) - f_R(x)|\mathrm{d}x \le A_i|h^i| \quad \forall\, R > 1 + |h^i|.$$

而由$f_R(x) = f(x)\mathbf{1}_{B(0,3R)}(x)$ 易见对任意$R > 1 + |h^i|$有

$$\mathbf{1}_{B(0,R)}|f(x + h^i\mathbf{e}_i) - f(x)| = \mathbf{1}_{B(0,R)}|f_R(x + h^i\mathbf{e}_i) - f_R(x)| \quad \forall\, x \in \mathbb{R}^d.$$

所以得到

$$\int_{B(0,R)} |f(x + h^i\mathbf{e}_i) - f(x)|\mathrm{d}x \le A_i|h^i| \quad \forall\, R > 1 + |h^i|.$$

令$R \to \infty$得到

$$\int_{\mathbb{R}^d} |f(x + h^i\mathbf{e}_i) - f(x)|\mathrm{d}x = \lim_{R\to\infty} \int_{B(0,R)} |f(x + h^i\mathbf{e}_i) - f(x)|\mathrm{d}x \le A_i|h^i|.$$

现在对任意$h \in \mathbb{R}^d \setminus \{0\}$, 令

$$H_1 = h = \sum_{j=1}^d h^j\mathbf{e}_j, \quad H_i = \sum_{j=i}^d h^j\mathbf{e}_j, \quad i = 1, 2, ..., d, \quad H_{d+1} = 0.$$

则有

$$f(x + h) - f(x) = f(x + H_1) - f(x + H_{d+1}) = \sum_{i=1}^d \big(f(x + H_i) - f(x + H_{i+1})\big),$$

$$\Longrightarrow \quad \big(\text{注意 } H_i = H_{i+1} + h^i\mathbf{e}_i,\ i = 1, 2, ..., d\big)$$

$$|f(x+h) - f(x)| \le \sum_{i=1}^{d} \big|f(x + H_i) - f(x + H_{i+1})\big|$$

$$= \sum_{i=1}^{d} \big|f(x + H_{i+1} + h^i\mathbf{e}_i) - f(x + H_{i+1})\big|$$

$$\Longrightarrow \quad (\text{由平移不变性})$$

$$\int_{\mathbb{R}^d} |f(x+h) - f(x)|\mathrm{d}x \le \sum_{i=1}^{d} \int_{\mathbb{R}^d} \big|f(x + H_{i+1} + h^i\mathbf{e}_i) - f(x + H_{i+1})\big|\mathrm{d}x$$

$$= \sum_{i=1}^{d} \int_{\mathbb{R}^d} \big|f(x + h^i\mathbf{e}_i) - f(x)\big|\mathrm{d}x \le \sum_{i=1}^{d} A_i|h^i|$$

$$\le \Big(\sum_{i=1}^{d} A_i^2\Big)^{1/2}\Big(\sum_{i=1}^{d} |h^i|^2\Big)^{1/2} = \Big(\sum_{i=1}^{d} A_i^2\Big)^{1/2}|h|.$$

所以

$$\frac{1}{|h|}\int_{\mathbb{R}^d} |f(x+h) - f(x)|\mathrm{d}x \le \Big(\sum_{i=1}^{d} A_i^2\Big)^{1/2}.$$

所以

$$\sup_{0\neq h\in\mathbb{R}^d} \frac{1}{|h|}\int_{\mathbb{R}^d} |f(x+h) - f(x)|\mathrm{d}x \le \Big(\sum_{i=1}^{d} A_i^2\Big)^{1/2} < \infty$$

所以$f \in \mathrm{BV}(\mathbb{R}^d)$. □